Physics Worksheet Solutions

Light & Kinematics – Grade 10

Academic Year 2025-2026

Exercise 1: Motion Recording Analysis

1. The movement of the mobile is rectilinear. Justify.
The successive positions $A_0, A_1, A_2, \dots$ recorded by the mobile form a straight line. Therefore, the trajectory is a straight line, making the movement rectilinear.

2. The movement of the mobile is accelerated. Justify.
For consecutive and equal time intervals ($\tau = 40$ ms), the distance covered by the mobile increases: $A_0A_1(1.2\text{ cm}) < A_1A_2(2.0\text{ cm}) < A_2A_3(2.8\text{ cm}) < A_3A_4(3.6\text{ cm}) < A_4A_5(4.4\text{ cm})$. Because the distance traveled in equal time intervals increases, the speed increases, meaning the movement is accelerated.

3. What is the abscissa of the mobile at $t_0 = 0$ and at $t_1 = 40$ ms?
Assumption: Assuming the origin O ($x = 0$) coincides with the starting point A_0.

At $t_0 = 0$: $x_0 = 0$ cm.
At $t_1 = 40$ ms: $x_1 = A_0A_1 = 1.2$ cm.

4.1. Calculate the instantaneous speeds V_2 and V_3.

- $V_2 = \frac{A_1A_3}{2\tau} = \frac{A_1A_2+A_2A_3}{2\times0.04} = \frac{2+2.8}{0.08} = \frac{4.8\text{ cm}}{0.08\text{ s}} = 60$ cm/s $= 0.6$ m/s.
- $V_3 = \frac{A_2A_4}{2\tau} = \frac{A_2A_3+A_3A_4}{2\times0.04} = \frac{2.8+3.6}{0.08} = \frac{6.4\text{ cm}}{0.08\text{ s}} = 80$ cm/s $= 0.8$ m/s.

4.2. Determine the characteristics of the velocity vector $\vec{v}_1$.

- **Point of application:** Position A_1
- **Direction:** Horizontal (along the trajectory)
- **Sense:** Direction of motion (Positive direction $+i$)
- **Magnitude:** $V_1 = 0.4$ m/s

4.3. Represent using the scale: 1 cm $\leftrightarrow 0.2$ m/s the vector $\vec{v}_1$.
Length of the vector on paper = $\frac{0.4\text{ m/s}}{0.2\text{ m/s/cm}} = 2$ cm.
(Draw a 2 cm arrow starting from A_1 pointing to the right).

5. Calculate the instantaneous accelerations a_2 and a_4.

- $a_2 = \frac{V_3-V_1}{2\tau} = \frac{0.8-0.4}{2\times0.04} = \frac{0.4}{0.08} = 5$ m/s^2.
- $a_4 = \frac{V_5-V_3}{2\tau} = \frac{1.2-0.8}{0.08} = \frac{0.4}{0.08} = 5$ m/s^2.

6. Determine the characteristics of the acceleration vector $\vec{a}_2$.

- **Point of application:** Position A_2

- **Direction:** Horizontal (along the trajectory)
- **Sense:** Same as the direction of motion ($+i$), since motion is accelerated.
- **Magnitude:** $a_2 = 5$ m/s^2

7. Represent using the scale: 1 **cm** $\leftrightarrow 0.2$ **m/s**^{2}**, the vector** $\vec{a}_2$**.**
Length of the vector on paper = $\frac{5 \text{ m/s}^2}{0.2 \text{ m/s}^2/\text{cm}} = 25$ cm.
(Note: A 25 cm vector is very long for a standard page. This is the mathematical result according to the scale provided in the prompt).

8. Calculate V_6 **knowing the acceleration is constant.**
Since acceleration $a = 5$ m/s^2 is constant:

$$V_6 = V_5 + a \times \tau = 1.2 + (5 \times 0.04) = 1.2 + 0.2 = 1.4 \text{ m/s.}$$

Exercise 2: Motion Recording Analysis

1. Indicate the positions occupied by particle (A) at t_0, t_1, t_2, t_3, t_4**.**
Using the scale given (2 cm per tick mark), and given the origin (O) is located exactly at position A_1:

- x_0 (at t_0): 4 intervals to the left of origin = $-4 \times 2 = -8$ cm.
- x_1 (at t_1): At the origin = 0 cm.
- x_2 (at t_2): 3 intervals right of origin = $+3 \times 2 = +6$ cm.
- x_3 (at t_3): 6 cm $+ (2$ intervals $\times$ 2 cm$) = +10$ cm.
- x_4 (at t_4): 10 cm $+ (1$ interval $\times$ 2 cm$) = +12$ cm.

2. What is the displacement of (A) during $\Delta t = t_1 - t_0$**?**
$\Delta x = x_1 - x_0 = 0 - (-8) = +8$ cm.

3. Calculate the average speed $V_{0,4}$ **between** t_0 **and** t_4**.**
$V_{0,4} = \frac{x_4 - x_0}{t_4 - t_0} = \frac{12-(-8)}{4 \times 0.04} = \frac{20 \text{ cm}}{0.16 \text{ s}} = 125$ cm/s $= 1.25$ m/s.

4. Calculate the instantaneous speeds $V_1, V_2,$ **and** V_3**.**

- $V_1 = \frac{x_2 - x_0}{2\tau} = \frac{6-(-8)}{0.08} = \frac{14 \text{ cm}}{0.08 \text{ s}} = 175$ cm/s $= 1.75$ m/s.
- $V_2 = \frac{x_3 - x_1}{2\tau} = \frac{10-0}{0.08} = \frac{10 \text{ cm}}{0.08 \text{ s}} = 125$ cm/s $= 1.25$ m/s.
- $V_3 = \frac{x_4 - x_2}{2\tau} = \frac{12-6}{0.08} = \frac{6 \text{ cm}}{0.08 \text{ s}} = 75$ cm/s $= 0.75$ m/s.

5. Determine the characteristics of the velocity vector $\vec{v}_2$**.**

- **Point of application:** Position A_2
- **Direction:** Horizontal (x-axis)
- **Sense:** Positive direction ($+i$)

– **Magnitude:** 1.25 m/s

6. Represent using the scale 1 cm $\leftrightarrow 1.25$ m/s, the velocity vector $\vec{v}_2$.
Length = $\frac{1.25}{1.25} = 1$ cm directed to the right from A_2.

7. Calculate the algebraic value of the acceleration vector $\vec{a}_2$.

$a_2 = \frac{V_3 - V_1}{2\tau} = \frac{0.75-1.75}{0.08} = \frac{-1.0 \text{ m/s}}{0.08 \text{ s}} = -12.5$ m/s^2.

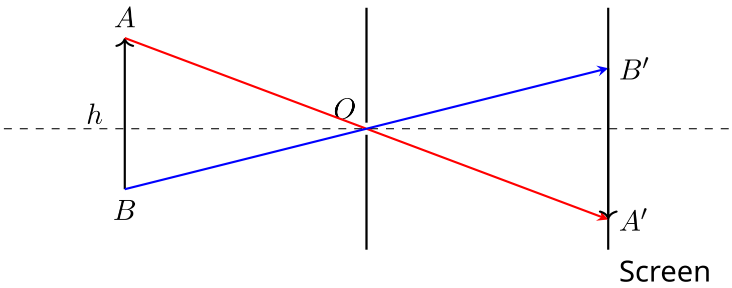
8. Write the expression of $\vec{a}_2$.
$\vec{a}_2 = -12.5\,\vec{i}$

9. Represent using the scale 1 cm $\leftrightarrow 10$ m/s^2, the vector $\vec{a}_2$.
Length = $\frac{12.5}{10} = 1.25$ cm. Vector drawn starting from A_2 pointing to the **left** (negative direction).

10. What's the nature of motion of (A)?
Since the velocity vector ($\vec{v}_2$) is positive and the acceleration vector ($\vec{a}_2$) is negative, their dot product is negative ($\vec{v}_2 \cdot \vec{a}_2 < 0$). Thus, the motion is a **rectilinear decelerated motion** (specifically, uniformly decelerated).

Exercise 3: Rectilinear propagation of light

1. Redraw the figure and trace the light rays.
Construction Steps: Draw a straight line from the top of the object (A) through the pinhole (O) to hit the bottom of the screen (producing A'). Draw a straight line from the bottom of the object (B) through O to hit the top of the screen (producing B'). The image $A'B'$ is inverted.



2. Determine the distance d separating the object from the black chamber.
By applying Thales' theorem (similar triangles formed by rays crossing at the pinhole O):

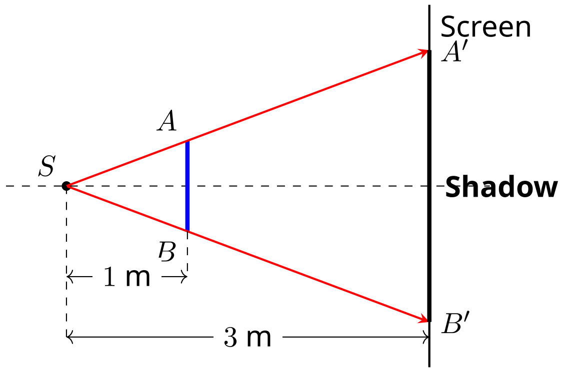
$$\frac{h'}{h} = \frac{D}{d}$$
$$\frac{10 \text{ cm}}{100 \text{ cm}} = \frac{50 \text{ cm}}{d}$$
$$\frac{1}{10} = \frac{50}{d} \implies d = 50 \times 10 = 500 \text{ cm (or 5 m).}$$

The object is placed at a distance $d = 500$ cm from the black chamber.

Exercise 4: Shadow of an object

1. Trace the light rays to obtain the shadow.

Construction Steps: Draw straight lines starting from the point source through the upper and lower tips of the object until they reach the screen. The gap between the intersections on the screen is the shadow.



2. Calculate the size of the shadow.

Using the property of similar triangles (Thales' theorem) between the source, the object, and the screen:

$$\frac{H_{\text{shadow}}}{h_{\text{object}}} = \frac{D_{\text{source}\to\text{screen}}}{d_{\text{source}\to\text{object}}}$$
$$\frac{H}{15\text{ cm}} = \frac{3\text{ m}}{1\text{ m}}$$
$$H = 15\text{ cm} \times 3 = 45\text{ cm}.$$

The size of the shadow is 45 cm.

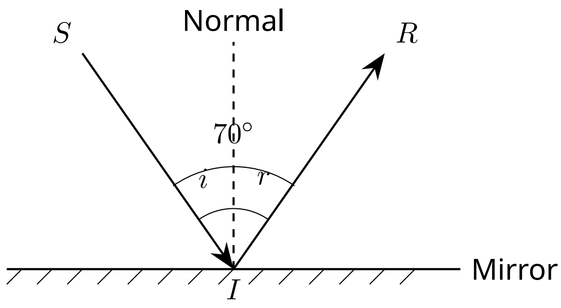
3. To increase the size of the shadow, should we get the source closer or further from the screen?

To increase the size of the shadow, we must increase the magnification ratio D/d. Moving the source **closer to the object** decreases the denominator d, which substantially increases the size of the shadow. Hence, the source should be moved **closer** to the object (and subsequently closer to the screen if the screen is fixed).

Exercise 5: Reflection of light

1. Redraw the figure and trace the normal.

Construction Steps: Draw a dashed line perpendicular ($90°$) to the plane mirror precisely at the point of incidence I.



2. Calculate the angle of incidence i. Deduce the reflection angle r.

The total angle between the incident ray and the reflected ray is $70°$. According to the law of reflection, the angle of incidence equals the angle of reflection ($i = r$).

$i + r = 70° \implies 2i = 70° \implies i = 35°$.
By deduction, the reflection angle is $r = 35°$.